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8.2.5 Preconditioning

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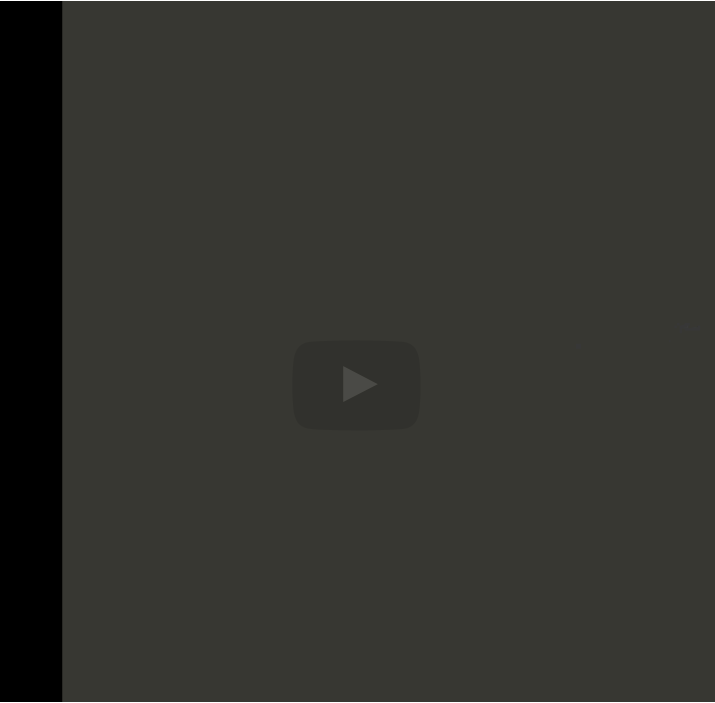
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matter of fact, it's a matter of replacing your search direction by M inverse times the residual. And you have the same net effect, where in the end, you end up with the vector x that you want. However, there remains a problem. And that is this matrix may not be symmetric positive-definite. And all of the discussion we've had so far is for symmetric positive definite matrices. So then either we need to generalize our

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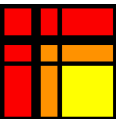
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Reading assignment

1/1 point (graded)



Read Unit 8.2.5 of the notes. [\[LINK\]](#)

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[Ill conditioning: oscillations drawn by Professor](#)

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At around 0:46, Robert explains the problems of ill conditioning and how the...

Homework Question

1/1 point (graded)
Show that the following algorithms

Given : $A, b, x^{(0)},$ $M = LL^T$ $\tilde{A} = L^{-1}AL^{-T}$ $\tilde{b} = L^{-1}b$ $\tilde{x}^{(0)} = L^T x^{(0)}$ $\tilde{r}^{(0)} := \tilde{b} - \tilde{A}\tilde{x}^{(0)}$ $k := 0$ while $r^{(k)} \neq 0$ $\tilde{p}^{(k)} := \tilde{r}^{(k)}$ $\tilde{q}^{(k)} := \tilde{A}\tilde{p}^{(k)}$ $\tilde{\alpha}_k := \frac{\tilde{p}^{(k)T}\tilde{r}^{(k)}}{\tilde{p}^{(k)T}\tilde{q}^{(k)}}$ $\tilde{x}^{(k+1)} := \tilde{x}^{(k)} + \tilde{\alpha}_k\tilde{p}^{(k)}$ $\tilde{r}^{(k+1)} := \tilde{r}^{(k)} - \tilde{\alpha}_k\tilde{q}^{(k)}$ $x^{(k+1)} = L^{-T}\tilde{x}^{(k+1)}$ $k := k + 1$ endwhile	Given : $A, b, x^{(0)}, M$ $r^{(0)} := b - Ax^{(0)}$ $k := 0$ while $r^{(k)} \neq 0$ $p^{(k)} := M^{-1}r^{(k)}$ $q^{(k)} := Ap^{(k)}$ $\alpha_k := \frac{p^{(k)T}r^{(k)}}{p^{(k)T}q^{(k)}}$ $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$ $r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)}$ $k := k + 1$ endwhile
--	--

compute the same values for $x^{(k)}$.

☒ done



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